

ANGELOS ZACHARIA

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EDUCATION

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|---|--|
| Norwegian University of Science and Technology (NTNU)
<i>Doctorate of Philosophy in Engineering Cybernetics</i>
· Supervisor: Prof. Kostas Alexis | Trondheim, Norway
Jan. 2024 - Present |
| National Technical University of Athens (NTUA)
<i>Master of Science in Automatic Control Systems and Robotics</i>
9.7/10 - Distinction (Ranked 1st in the class)
· Thesis: “Event-Triggered Distributed Control for Optimal Consensus of Unknown Nonlinear Agents in Normal Form”
· Supervisor: Asst. Prof. Haris E. Psillakis | Athens, Greece
Oct. 2020 - Jun. 2022 |
| University of Cyprus (UCY)
<i>Bachelor of Science in Electrical Engineering</i>
8.27/10 - Very Good (Ranked 3rd in the class)
· Thesis: “Sensor Fault Accommodation for UAV Altitude Control: Virtual Sensor Approach”
· Supervisor: Prof. Marios M. Polycarpou | Nicosia, Cyprus
Sept. 2015 - Jul. 2019 |

WORK EXPERIENCE

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| Autonomous Robots Lab, NTNU
<i>PhD Candidate</i>
· Conducting research on heterogeneous robotic systems-of-systems for rapid exploration.
· Part of the Horizon Europe project: “SYNERGIZE: A novel integrated SYstem of Systems streNgthening tEchnical and logistical capacities to ensure better Response to emerGencies by synergIS-tically addrEssing FRs capability gap” | Trondheim, Norway
Jan. 2024 - Present |
| KIOS Research & Innovation CoE, University of Cyprus
<i>Research Engineer I</i>
· Developed cooperative control algorithms for multi-drone 3D inspection.
· Conducted simulations and real-world experiments using multi-drone systems.
· Part of the project: “GLIMPSE: Intelligent Multi-drone Emergency Response System”
· Member of Emergency Response Technologies team at KIOS CoE. | Nicosia, Cyprus
Sept. 2022 - Dec. 2023 |
| <i>Research Engineer I</i>
· Developed “ICARUS: Power dIstribution network inspeCtion plAtfoRm Using UAVS”.
· Conducted real-world field tests and demos of the integrated platform.
· Part of the funded project by the Electricity Authority of Cyprus
· Member of Emergency Response Technologies team at KIOS CoE. | Jan. 2020 - Oct. 2020 |
| YTM Stavrides Ltd
<i>Assistant Engineer</i>
· Designed electrical circuits for electrical installations in AutoCAD.
· Worked on tender documents and was responsible for measuring the requirements of circuits, control panels, and structured cabling. | Nicosia, Cyprus
Jun. 2016/17 - Aug. 2016/17 |

Cyprus National Guard
Second Lieutenant of Artillery Division

Larnaca, Cyprus
Sept. 2013 - Sept. 2015

- Ranked among the top 150 recruits out of 5,000 in both intellectual and fitness tests and underwent a 4-month training at the Reserve Officers School of Artillery in Nea Peramos, Greece.

PUBLICATIONS

Journals

Zacharia A, Dharmadhikari M, Alexis K. Collaborative Exploration with a Marsupial Ground-Aerial Robot Team through Task-Driven Map Compression. *IEEE Robotics and Automation Letters*. 2025.

Cao M, Nguyen TM, Yuan S, Anastasiou A, **Zacharia A**, Papaioannou S, Kolios P, et al. Cooperative Aerial Robot Inspection Challenge: A Benchmark for Heterogeneous Multi-Uncrewed-Aerial-Vehicle Planning and Lessons Learned. *IEEE Robotics & Automation Magazine*. 2025.

Conferences

Anastasiou A, **Zacharia A**, Papaioannou S, Kolios P, Panayiotou CG, Polycarpou MM. Automated real-time inspection in indoor and outdoor 3d environments with cooperative aerial robots. *IEEE International Conference on Unmanned Aircraft Systems (ICUAS)*. 2024.

Zacharia A, Papaioannou S, Kolios P, Panayiotou C. Distributed Control for 3D Inspection using Multi-UAV Systems. *IEEE Mediterranean Conference on Control and Automation (MED)*. 2023.

Savva A, **Zacharia A**, Makrigiorgis R, Anastasiou A, Kyrkou C, Kolios P, Panayiotou C, Theocharides T. ICARUS: Automatic autonomous power infrastructure inspection with UAVs. *IEEE International Conference on Unmanned Aircraft Systems (ICUAS)*. 2021.

Souli N, Makrigiorgis R, Anastasiou A, **Zacharia A**, Petrides P, Lazanas A, Valianti P, Kolios P, Ellinas G. Horizonblock: Implementation of an autonomous counter-drone system. *IEEE International Conference on Unmanned Aircraft Systems (ICUAS)*. 2020.

ACHIEVEMENTS

**Cooperative Aerial Robots Inspection Challenge,
62nd IEEE Conference on Decision and Control**

First Place, Singapore 2023

- Designed a cooperative inspection algorithm utilizing multi-UAV systems for infrastructure inspection.

2nd CASS Student Design Competition, IEEE Circuits and Systems Society

Finalist at the world finals held in Florence, Italy in May 2018.

- Designed a versatile and low-cost, two layer PCB for soil quality monitoring.
(Project: “A Versatile Electronic Interface for Soil Quality Monitoring”)

TECHNICAL SKILLS

Programming Languages: C, C++, Python

Simulation and Modeling: Gazebo, RViz, MATLAB, Simulink

Operating Systems and Tools: Windows, Linux, ROS, Fusion 360

Technical Documentation: LaTeX, Microsoft Office

Languages: Greek (Native), Romanian (Fluent), English (Fluent)